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GRADES 7–10

# EMERGENCY SUB PLANS PACK

## for DIGITAL TECHNOLOGIES

★ *Keep learning on track—even when plans change.* ★

VOLUME 2


- NO-PREP ACTIVITIES
- READY-TO-USE RESOURCES
- PERFECT FOR EMERGENCY LESSONS
- GREAT FOR SUBSTITUTE TEACHERS
- ENGAGING DIGITAL TECH TASKS
- PRINT OR DIGITAL USE

YOUR *LIFESAVER* IN THE CLASSROOM

SAVE TIME | REDUCE STRESS | KEEP STUDENTS

MADE BY TEACHERS FOR TEACHERS

BEFORE YOU NEED IT

# Set this up while you are well

FOR THE REGULAR TEACHER

15 MINUTES

You are reading this because you are **well**. Spend fifteen minutes now and this pack will run itself on a day when you are not.

### The one job you have today

Print pages 3–6 of this pack and leave them in a clearly labeled folder where your substitute teacher will find them — top drawer, on the desk, or with the front office. Everything a substitute needs is in those four pages.

You do not need to print the lessons in advance. The substitute teacher chooses one on the day and photocopies it.

## SETTING IT UP · FIFTEEN MINUTES, ONCE

1. **Print the whole pack once.** Put it in a display folder or bind it. Write your name and room number on the cover.
2. **Print pages 3–6 a second time** and clip them to the front. These are the pages a substitute teacher reads first.
3. **Tell your department head where it lives.** On a genuine emergency day you will not be answering your phone, so somebody else needs to know.
4. **Cross off the lessons you have already used.** There is a tracker on page 4. Two years of substitute days will not repeat if you keep it current.
5. **Optional: leave a class list** and a note about any students who need specific support, seating or a reader.

### Why these lessons are unplugged

On a substitute day the laptop cart is often locked, the login server is down, or nobody knows the room password. Every lesson here runs on paper.

Each lesson also has an **If the computers are working** note, so the same sheet still works when technology cooperates.

### Why they are not busy work

Every lesson targets a genuine Digital Technologies concept and produces marked evidence you can put in a folder.

If you are away for a week, running four of these in sequence is a defensible unit of work, not a holding pattern.

### A word about the answer keys

Every answer key sits on the **last page of its lesson**. If you hand a lesson to a substitute teacher, hand them the answer key too — it is the single thing that lets a non-specialist mark with confidence and answer a student who says *"but why?"*

### This is Volume Two

Volume 2 stands completely on its own. Nothing in these ten lessons repeats Volume 1 — there is no binary conversion, no trace table, no flowchart and no password ranking here. If you own both, you have twenty substitute days that never overlap.

SUBSTITUTE TEACHER · START HERE

# You do not need to know this subject

READ THIS FIRST

2 MINUTES

You have walked into a Digital Technologies class. You may have never taught the subject. That is completely fine — this pack was written for you.

### Read this and you are ready

**You do not need to know how to code.** Every lesson is a paper worksheet with a full answer key. Your job is to hand it out, keep the room settled, and check work as students finish. You are not expected to teach the content.

DO THESE FOUR THINGS

1. **Pick one lesson** from the table on the next page. Any of them works for any class from Grade 7 to Grade 10. If you cannot decide, use Lesson 04.
2. **Photocopy the student pages only.** Each lesson tells you exactly which page numbers to copy. One copy per student.
3. **Keep the teacher page and the answer key.** Do not photocopy those.
4. **Read the teacher page once** before the bell. It takes about two minutes and includes the exact words you can say to start the lesson.

### If a student says “we have done this”

Ask them to prove it by completing the extension section at the end. If they genuinely have, switch them to a different lesson from the table — the pack has ten, and they will not have done them all.

### If you finish early

Every lesson has an extension task on its final student page. Point students there. There is also a discussion prompt on the teacher page you can run as a whole class for the last five minutes.

### What to leave for the regular teacher

Collect the completed sheets. Fill in the handover note on the last page of this pack — it takes ninety seconds and tells the regular teacher which lesson you used, how far the class got, and anything that happened.

Leave both on the desk.

SUBSTITUTE TEACHER · STEP ONE

# Choose your lesson

PICK ANY ONE

10 OPTIONS

Every lesson is standalone. There is no order and no prerequisite — pick whichever suits the class in front of you.

| # | LESSON | WHAT IT COVERS | DIFFICULTY | MINS | STUDENT PAGES |
|---|---|---|---|---|---|
| **01** | Clean the Data | Data errors and validation | ★★ | 60 | 7–9 |
| **02** | Charts That Lie | Data visualization and honesty | ★ | 55 | 12–14 |
| **03** | Databases by Hand | Records, fields, keys and queries | ★★ | 60 | 17–19 |
| **04** | Input, Process, Output | Systems thinking and sensors | ★ | 55 | 22–24 |
| **05** | Making It Smaller | Compression, lossy and lossless | ★★ | 60 | 27–29 |
| **06** | Secret Writing | Encryption, keys and keyspace | ★★ | 60 | 32–34 |
| **07** | Design for a Real User | User needs, criteria and interfaces | ★ | 55 | 37–39 |
| **08** | Break It Into Blocks | Functions, parameters and scope | ★★★ | 60 | 42–44 |
| **09** | Planning a Project | Task breakdown and scheduling | ★ | 55 | 47–49 |
| **10** | What a Free App Actually Costs | Data, consent and business models | ★ | 55 | 52–54 |

### Easiest to supervise

**02, 04 and 10.** Reading, judgment and argument. Students have opinions, so the room stays busy and nobody gets stuck.

### Best for a quiet, working class

**03, 05 and 08.** Puzzle-like with definite answers. Strong students will race; that is what the extensions are for.

### Best for a restless class

**01, 07 and 09.** Hunting for faults, sketching and shading a chart. Noisier, but productively so.

### Used-lesson tracker · for the regular teacher

| LESSON | DATE USED | CLASS | NOTES |
|---|---|---|---|
| 01 — Cleaning data | | | |
| 02 — Misleading charts | | | |
| 03 — Databases | | | |
| 04 — Systems and sensors | | | |
| 05 — Compression | | | |
| 06 — Encryption | | | |
| 07 — Design and interfaces | | | |
| 08 — Functions | | | |
| 09 — Project planning | | | |
| 10 — Free apps and data | | | |

SUBSTITUTE TEACHER · STEP TWO

# The first ten minutes

WHAT TO SAY

SCRIPT

The first four minutes decide the next fifty. Here is a script that works even if you have never met this class.

### SAY THIS AT THE DOOR

> *“Come in, sit down, and leave your bags off the desks. You will not need a computer today. Everything is on paper.”*

### THEN, ONCE THEY ARE SEATED

> *“Your teacher is away. My name is ______. We are doing a proper Digital Technologies lesson today, not a free period. There is a worksheet, it gets collected at the end, and your teacher will see it.”*

**Why that last sentence matters**

The single most useful thing you can say is that the work will be seen by their regular teacher. It is also true — there is a handover note at the back of this pack.

### THE SHAPE OF EVERY LESSON IN THIS PACK

| TIME | WHAT HAPPENS | WHAT YOU DO |
|---|---|---|
| 0–5 | Settle, hand out sheets | Use the script above |
| 5–10 | Read the opening page together | Ask one student to read it aloud |
| 10–45 | Students work through the tasks | Circulate. Check answers against the key |
| 45–55 | Extension task for those who finish | Point them to the last student page |
| 55–60 | Collect sheets, pack up | Fill in the handover note |

### If a student refuses to work

Offer the smaller ask: *“Just do question one and show me.”* Most refusals are about not knowing where to start, not defiance.

Note it on the handover sheet. Do not escalate a single lesson into a battle you will not be there to finish.

### If the whole class claims they cannot do it

Do the first question together on the board. Every lesson has a worked example on the first student page.

If it is genuinely too hard for the grade level, switch to Lesson 04 or Lesson 10 — both need no prior knowledge at all.

### Photocopying reminder

Student pages only. Each lesson’s teacher page states the exact page range. The **answer key is always the last page** of the lesson — keep it.

LESSON 01 · TEACHER BRIEFING

# Clean the Data

READ BEFORE THE BELL

2 MINUTES

Students find eight planted faults in a small dataset, write validation rules that would have stopped them, then answer questions the data can and cannot support. It is the least glamorous and most useful skill in the subject.

### You do not need to know this subject

Every fault in the table is visible to anyone reading carefully — a missing number, an impossible date, a name typed twice. The key names all eight and explains each one.

### Photocopy these pages only

**Pages 7–9**

One per student. **Do not photocopy this page or the answer key.**

### What students need

A pen. A calculator is useful for one question but not essential.

## WHAT TO SAY TO START

> *"Every organization you will ever work for has a database with junk in it. Today you are going to find the junk, work out how it got there, and write the rule that would have stopped it at the door."*

## HOW THE LESSON RUNS

| TIME | WHAT HAPPENS | WHAT YOU DO |
|---|---|---|
| 0–5 | Hand out sheets, settle | Use the door script on page 5 |
| 5–12 | Read the panel and worked example together | Point out the four fault types |
| 12–28 | Task 1.1 — find the eight faults | There are exactly eight. Key lists them |
| 28–40 | Task 1.2 — write validation rules | Judgment, not arithmetic |
| 40–50 | Task 1.3 and 1.4 — using the cleaned data | One calculation. Key has the number |
| 50–60 | Task 1.5, collect sheets | Discussion prompt below |

### Questions students will ask

**"Is a spelling mistake really an error?"**
If it stops a computer matching two records, yes. `ben carter` and `Ben Carter` are different strings to a machine.

**"Can I just delete the bad rows?"**
Sometimes. But deleting Eve because her score is missing throws away everything else she has. That trade-off is Task 1.4.

**"How many faults are there?"**
Eight, in eight different rows. Row 10 is completely clean, which students find suspicious.

### If they finish early

Task 1.5 asks how you would check 40,000 rows you cannot read. Strong students can then design a one-page entry form that makes seven of the eight faults impossible to type in the first place.

### If the computers are working

Students can type the table into a spreadsheet and use conditional formatting or a filter to surface the faults. Ask them to find them on paper first, then see whether the machine agrees.

### Curriculum focus · for the regular teacher's records

Acquiring, validating and interpreting data; data types; the effect of data quality on the information produced. Discussion prompt for the last five minutes: *"A report says the Grade 9 average is 64. It was calculated from this table. Is the report wrong, or just unlucky?"*

LESSON 01 · CLEAN THE DATA

# Garbage in, garbage out

UNDERSTAND

16 MIN

NAME

CLASS

DATE

## Data is not information until somebody cleans it

A computer will happily add up a column that contains a score of 105, a blank, and the word `Nine`. It will produce a confident average and it will be wrong.

Real datasets arrive broken. Learning to see the breakage is the first job of anyone who works with data.

### Missing

A field left empty. Not the same as a zero — a zero is a score, a blank is a question.

### Impossible

A value outside what reality allows: a score of 105, a date of June 31.

### Wrong type

Words where numbers belong. `Nine` cannot be added up.

### Inconsistent

The same thing written two ways: 05/22/2025 and 05-22-2025.

## Task 1.1 · Find the eight faults

DETECTIVE

Every row below except one has exactly one fault. Circle it, then write which of the four types it is. There is one extra type to watch for: the same person entered twice.

| # | STUDENT | GRADE | SCORE | DATE JOINED | EMAIL | FAULT TYPE |
|---|---|---|---|---|---|---|
| 1 | Aisha Rahman | 9 | 72 | 05/03/2025 | aisha.r@school.edu | |
| 2 | ben carter | 9 | 68 | 05/03/2025 | ben.c@school.edu | |
| 3 | Chloe Nguyen | 9 | 105 | 05/12/2025 | chloe.n@school.edu | |
| 4 | Dan Oliveira | Nine | 54 | 05/12/2025 | dan.o@school.edu | |
| 5 | Aisha Rahman | 9 | 72 | 05/03/2025 | aisha.r@school.edu | |
| 6 | Eve Larsson | 10 | | 05/19/2025 | eve.l@school.edu | |
| 7 | Farid Haddad | 10 | 61 | 06/31/2025 | farid.h@school.edu | |
| 8 | Gia Romano | 10 | 88 | 05/22/2025 | gia.r.school.edu | |
| 9 | Hana Kim | 10 | 47 | 05-22-2025 | hana.k@school.edu | |
| 10 | Isaac Bell | 10 | 79 | 05/25/2025 | isaac.b@school.edu | |

Which row is clean? ________ Which fault would be hardest for a computer to notice on its own?

________________

LESSON 01 · CLEAN THE DATA

# Rules that stop bad data

KEY SKILL

20 MIN

## Task 1.2 · Write the validation rule

PROVE

For each field, write a rule strict enough to reject the fault you found, but not so strict that it rejects a real student. Plain English is fine.

| FIELD | A FAULT IT MUST CATCH | THE RULE |
|---|---|---|
| Grade | Nine | |
| Score | 105 | |
| Date joined | 06/31/2025 | |
| Email | gia.r.school.edu | |
| Student | the same person twice | |

One of these five rules is much harder to write than the other four. Which one, and why?

## Task 1.3 · Now use the clean data

CALCULATE

Cross out the duplicate row and the impossible score. Work only with what is left.

| # | QUESTION | WORKING | ANSWER |
|---|---|---|---|
| a | How many usable records are left? | | |
| b | What is the mean score for Grade 9? | | |
| c | Who has the highest score? | | |
| d | How many Grade 10 students have a score recorded? | | |

**e.** Eve has no score. Should she count as a zero in the Grade 10 average? Explain in one sentence.

LESSON 01 · CLEAN THE DATA

# One person, two records

STRETCH

16 MIN

## Task 1.4 · Are these the same person?

DECIDE

**Record A**

Nguyen, Chloe
Class 9C
chloe.n@school.edu
Last updated 05/12/2025

**Record B**

Chloe Nguyen
Class 9C
c.nguyen@school.edu
Last updated 08/04/2025

**a.** Are they the same person? What is your evidence?

**b.** Which record would you keep, and what would you copy across from the other one?

**c.** What single extra field would let a computer answer part a with certainty? ______________________

## Task 1.5 · Forty thousand rows

THINK

You are handed a spreadsheet with 40,000 rows. You cannot read it. Describe three checks you would run to find the faults without looking at every row.

1. ______________________
2. ______________________
3. ______________________

**The last word.** A computer never notices that data is wrong, only that it is missing or the wrong shape. Explain why that difference matters.

## Before you hand this in

CHECKPOINT

**Check what you managed today**

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

**One thing I now understand that I did not this morning**

Teacher initials ______________________

# Answer key

Clean the Data

NOT FOR STUDENTS

ANSWERS

## Task 1.1 · the eight faults

| ROW | THE FAULT | TYPE |
|---|---|---|
| 2 | ben carter in lower case | Inconsistent format |
| 3 | Mark of 105 | Impossible value |
| 4 | Grade recorded as Nine | Wrong type |
| 5 | Aisha Rahman entered a second time | Duplicate record |
| 6 | Score left blank | Missing value |
| 7 | 06/31/2025 — June has 30 days | Impossible value |
| 8 | Email has no @ | Malformed / wrong type |
| 9 | Date written with hyphens | Inconsistent format |

**Row 10 is clean.** The hardest fault for a computer to catch is the duplicate: both copies are perfectly valid on their own. Nothing about row 5 is wrong except that row 1 already exists.

## Task 1.2 · model rules

| FIELD | RULE |
|---|---|
| Grade | Must be a whole number, 7 to 10 |
| Score | Whole number, 0 to 100 |
| Date | A real calendar date, mm/dd/yyyy, not in the future |
| Email | Exactly one @, a dot after it, no spaces |
| Student | No two records may share the same student ID |

**The hard one is the last.** Every other rule can be checked by looking at one value. Finding a duplicate means comparing a value against every other record, and names are not unique anyway — which is why databases use an ID.

## Task 1.3 · the numbers

**a.** Eight usable records. Row 5 is deleted as a duplicate and row 3's score is unusable.

**b.** Grade 9 scores left are 72, 68 and 54. 194 ÷ 3 = **64.67**. Accept 64.7 or 65 with working shown.

**c. Gia Romano**, 88.

**d. Four** — Farid, Gia, Hana, Isaac. Eve has a record but no score.

**e.** No. A zero says she took the task and scored nothing. A blank says nobody knows. Turning one into the other invents data and drags the average down. Exclude her from the average and say so.

## Task 1.4 · the duplicate

**a.** Almost certainly the same person: same name reordered, same class, same school email pattern.

**b.** Keep B — it was updated three months later, so the newer email is more likely to work. Copy nothing across; A holds nothing B lacks. Accept keeping A if the student argues the older address is the official one, provided they justify it.

**c.** A **student ID**: one value, never reused, that belongs to a person rather than describing them. Accept 'unique ID', 'primary key' or 'student number'.

## Task 1.5 · checking 40,000 rows

Good answers name checks a machine can run: count the blanks in each column · sort each column and read the top and bottom values (outliers surface immediately) · check every value against its type · count how many times each ID appears · count the distinct spellings of a category.

**The last word:** a computer can test the *shape* of a value, not its truth. 68 is a perfectly valid score even if the student actually scored 86. Validation stops nonsense, not lies or typos that happen to be plausible.

LESSON 02 · TEACHER BRIEFING

# Charts That Lie

READ BEFORE THE BELL

2 MINUTES

Students meet the same three numbers drawn two ways, name six standard tricks, then repair a misleading chart. Highly visual and argumentative in a good way.

### You do not need to know this subject

Every answer is either a number read off a chart or a judgment about whether the picture matches the numbers. The key gives both.

### Photocopy these pages only

**Pages 12–14**

One per student. **Do not photocopy this page or the answer key.**

### What students need

A pen. A ruler is useful for Task 2.3 but not required.

## WHAT TO SAY TO START

> *"Nobody has to make up a number to lie to you with data. All they have to do is choose where the bottom of the chart goes. Today you will do it yourself, and then you will never unsee it."*

## HOW THE LESSON RUNS

| TIME | WHAT HAPPENS | WHAT YOU DO |
|---|---|---|
| 0–5 | Hand out sheets, settle | Use the door script on page 5 |
| 5–12 | Read the two charts together | Ask what the difference looks like before reading the axis |
| 12–24 | Task 2.1 — read the chart honestly | Answers are exact numbers |
| 24–36 | Task 2.2 — name the six tricks | Six answers, all in the key |
| 36–48 | Task 2.3 and 2.4 — fix it, then write both headlines | Expect argument. Good |
| 48–55 | Task 2.5, collect sheets | Discussion prompt below |

### Questions students will ask

**"Is it actually lying if the numbers are right?"**
That is the whole lesson. Nothing on the first chart is false, and it still makes you believe something untrue.

**"Should every chart start at zero?"**
Bar charts, yes — the bar length is the message. A line chart may start elsewhere if the axis is clearly labeled.

**"Which chart type do I use?"**
Bars compare separate things. Lines show change over time. Pie shows parts of one whole.

### If they finish early

Task 2.4 asks for an honest headline and a misleading one from the same data. Strong students can then find a real chart in an advertisement and name the technique it uses.

### If the computers are working

Students can build both versions in a spreadsheet by changing one axis setting. Watching the same data redraw itself is more convincing than any explanation.

### Curriculum focus · for the regular teacher's records

Representing and interpreting data; selecting appropriate visualizations; evaluating how data is used to persuade. Discussion prompt for the last five minutes: *"A chart and the table it came from contain exactly the same numbers. Why is the chart more persuasive — and is that a good thing?"*

# Same numbers, two pictures

UNDERSTAND

14 MIN

NAME

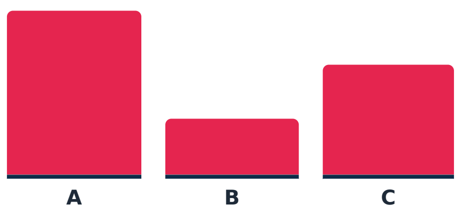
CLASS

DATE

## Nothing here is false

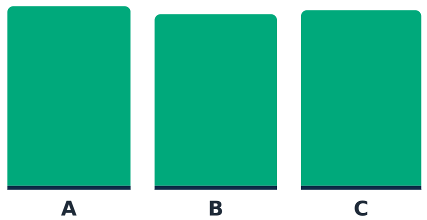
Three schools ran the same test. School A scored 82, School B scored 78, School C scored 80. Both charts below are drawn from those three numbers and neither contains an error.

### Chart 1 · as published




Vertical axis starts at 76

### Chart 2 · the same data




Vertical axis starts at 0

## Task 2.1 · Read it properly

READ

| # | QUESTION | ANSWER |
|---|---|---|
| a | From Chart 1 alone, School A looks roughly how many times better than School B? | |
| b | What is the actual difference between A and B, in points? | |
| c | What percentage of School A's score is that difference? | |
| d | Where does the axis start on Chart 1? | |
| e | Which chart would a school with 78 prefer to publish, and why? | |

**f.** Write one sentence that is *true* about these three schools and would still be true in a year's time.

LESSON 02 · CHARTS THAT LIE

# Six ways to bend a picture

KEY SKILL

20 MIN

## Task 2.2 · Name the trick

CLASSIFY

Match each description to its name. Write the letter in the last column.

| # | WHAT WAS DONE | TRICK |
|---|---|---|
| 1 | The vertical axis starts at 76, not 0 | |
| 2 | Only the three best months are shown | |
| 3 | The five slices of the pie add up to 140% | |
| 4 | Years run 2015, 2016, 2020, 2021 across the axis | |
| 5 | Bars are drawn as 3-D blocks tilted away from the reader | |
| 6 | The axis has no numbers on it at all | |

### The names

**A** Truncated axis
**B** Cherry-picked range
**C** Impossible total
**D** Uneven intervals
**E** Decorative distortion
**F** No scale

Two of these six make the chart wrong no matter how it is labeled. Which two? ______________________

## Task 2.3 · Choose the right chart

DECIDE

| # | WHAT YOU WANT TO SHOW | CHART TYPE | AXIS STARTS AT |
|---|---|---|---|
| a | How many students chose each of four electives | | |
| b | How the school population changed each year since 2015 | | |
| c | What share of the cafeteria’s takings came from drinks | | |
| d | Whether taller students have longer arms | | |

One of these four is the only one where a pie chart is honest. Which, and why?

______________________________

LESSON 02 · CHARTS THAT LIE

# Write it both ways

STRETCH

16 MIN

## Task 2.4 · The two headlines

WRITE

A school’s attendance over five terms: `91%, 89%, 88%, 90%, 92%`.

**a.** Write a headline that is honest.

**b.** Write a headline that is technically true and badly misleading.

**c.** Which two numbers did your misleading headline use, and which three did it ignore?

**d.** Describe the chart you would draw to support headline b.

## Task 2.5 · The last word

THINK

A chart and the table it came from contain exactly the same numbers. Explain in one sentence why people argue with the table but believe the chart.

## Before you hand this in

CHECKPOINT

**Check what you managed today**

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

**One thing I now understand that I did not this morning**

Teacher initials

# Answer key

Charts That Lie

NOT FOR STUDENTS

ANSWERS

## Task 2.1 · reading it properly

**a.** Roughly **three times** the height. Accept anything from twice to four times — the point is that it is nowhere near the truth.

**b. 4 points** (82 − 78).

**c.** 4 ÷ 82 = **4.9%**. Accept 'about 5%'.

**d. 76.**

**e.** Chart 2. The gap almost disappears when the bars are drawn from zero.

**f.** Look for a sentence that survives contact with the data: 'all three schools scored within five points of each other'. Reject anything with 'far ahead' in it.

## Task 2.2 · the six tricks

**1–A** truncated axis · **2–B** cherry-picked range · **3–C** impossible total · **4–D** uneven intervals · **5–E** decorative distortion · **6–F** no scale.

**The two that are wrong regardless of labeling: 3 and 4.** A truncated axis is defensible if it is labeled; slices totalling 140% and a timeline with missing years are broken whatever the caption says.

Students often argue 5 belongs there too. That is a good argument — a tilted 3-D bar cannot be read accurately even when honestly intended.

## Task 2.3 · choosing the chart

| # | CHART | AXIS | WHY |
|---|---|---|---|
| a | Bar chart | 0 | Four separate categories being compared by size |
| b | Line chart | Can start above 0 if labeled | One thing changing over time |
| c | Pie chart | n/a | Genuine parts of a single whole, adding to 100% |
| d | Scatter plot | n/a | Two measurements per person, looking for a relationship |

**Only c is honest as a pie.** The four electives in a look like parts of a whole but students may choose more than one, so the slices would not add up. That is exactly how impossible pies get published.

## Task 2.4 · what to look for

**Honest:** 'Attendance steady between 88% and 92% over five terms.'

**Misleading but true:** 'Attendance climbs for two straight terms' or 'attendance up 4 points from its low'.

**c.** The misleading version uses 88 and 92 and ignores 91, 89 and 90 — the three that show the movement is noise.

**d.** A line chart of terms 3, 4 and 5 only, with the axis starting at 87. Full credit for naming both the cherry-picked range and the truncated axis.

## Task 2.5 · the closing question

A table asks you to read and compare; a chart hands you a conclusion before you have read anything. The picture is processed faster than the numbers, so the impression arrives first and the checking arrives later — if at all.

Accept any answer reaching 'you see the shape before you read the numbers'.

LESSON 03 · TEACHER BRIEFING

# Databases by Hand

READ BEFORE THE BELL

2 MINUTES

Students run three queries over a library loans table with a pen, discover why one flat table repeats itself, and split it into two linked tables. This is relational database design without a computer.

### You do not need to know this subject

Every query answer is a list of rows you can count off the table, and all three are written out in the key. No software and no SQL is required.

### Photocopy these pages only

**Pages 17–19**

One per student. **Do not photocopy this page or the answer key.**

### What students need

A pen. A highlighter is useful for Task 3.2 but not required.

## WHAT TO SAY TO START

> *"A database is a table with rules. Today you are the database: I will give you three questions and you will answer them the way a machine does, one row at a time, without skipping."*

## HOW THE LESSON RUNS

| TIME | WHAT HAPPENS | WHAT YOU DO |
|---|---|---|
| 0–5 | Hand out sheets, settle | Use the door script on page 5 |
| 5–14 | Read the panel and the parts of a table | Get them to say record and field aloud |
| 14–24 | Task 3.1 — the primary key | Expect argument about names. Argument is good |
| 24–40 | Task 3.2 — run three queries | Exact answers. Grade as you circulate |
| 40–52 | Task 3.3 — split the table | The hardest and most valuable task |
| 52–60 | Task 3.4, collect sheets | Discussion prompt below |

### Questions students will ask

**"Can two people have the same name?"**
Yes, and that is precisely why a name cannot be the key. Ask how many Nguyens are in the school.

**"Why is the date written as Aug 5?"**
To keep the arithmetic simple. Today is August 12 for every question on the sheet.

**"Do I have to write out every row?"**
Yes. A database checks every row every time. Students who eyeball it get Task 3.2b wrong.

### If they finish early

Task 3.4 asks what breaks when two records share a key. Strong students can then add a third table for the books themselves and say which fields move into it.

### If the computers are working

Students can build the two-table version in any spreadsheet with two sheets, or in a database tool if the school has one. The paper version first, or the structure never gets thought about.

### Curriculum focus · for the regular teacher's records

Structuring data in tables; records, fields and keys; querying and sorting data; removing redundancy through related tables. Discussion prompt for the last five minutes: *"The library could keep everything in one big table. It would work. Why would nobody sensible do it?"*

LESSON 03 · DATABASES BY HAND

# Records, fields and keys

UNDERSTAND

14 MIN

NAME

CLASS

DATE

## A table with rules

A **record** is one row: everything known about one thing. A **field** is one column: the same fact about every thing.

A **primary key** is the field that identifies a record with no argument. It must be unique, it must never be empty, and it must never change.

## Task 3.1 · Choose the key

JUDGE

For each candidate field, check whether it could serve as the primary key for a table of students, and say what breaks if you use it.

| FIELD | COULD IT BE THE KEY? | WHAT BREAKS |
|---|---|---|
| Full name | | |
| Email address | | |
| Date of birth | | |
| Student ID | | |
| Phone number | | |

Two of the five are unique *today* but still make poor keys. Which two, and what do they have in common?

## The loans table · today is August 12

| LOAN | MEMBER | GRADE | TITLE | GENRE | DUE | RETURNED |
|---|---|---|---|---|---|---|
| L01 | Priya Shah | 9 | Ghost Boys | Fiction | Aug 5 | No |
| L02 | Priya Shah | 9 | Solar Cars | Non-fiction | Aug 19 | No |
| L03 | Tom Reilly | 10 | Nimona | Graphic | Aug 2 | Yes |
| L04 | Ana Silva | 9 | Ghost Boys | Fiction | Aug 8 | No |
| L05 | Ana Silva | 9 | The Rift | Fiction | Aug 26 | No |
| L06 | Jed Mullins | 8 | Solar Cars | Non-fiction | Aug 1 | No |
| L07 | Tom Reilly | 10 | Bird Atlas | Non-fiction | Aug 20 | No |
| L08 | Mia Kovac | 8 | Nimona | Graphic | Jul 30 | Yes |
| L09 | Ana Silva | 9 | Bird Atlas | Non-fiction | Aug 11 | No |
| L10 | Jed Mullins | 8 | The Rift | Fiction | Aug 22 | No |

# Be the database

KEY SKILL

20 MIN

## Task 3.2 · Run three queries

QUERY

Work through every row in order. Do not skip. Write the loan codes.

| # | THE QUERY | ROWS THAT MATCH | HOW MANY |
|---|---|---|---|
| a | Every loan that is overdue today | | |
| b | Every loan of a non-fiction book | | |
| c | Members with more than one book still out | | |
| d | Loans by Grade 9 members that are not overdue | | |

**e.** Query c is harder than the other three. What extra thing did you have to do?

______________________________

## Task 3.3 · The same thing, written twice

REDESIGN

‘Priya Shah’ and ‘Grade 9’ appear in this table more than once. That is **redundancy**, and it causes three specific problems.

| PROBLEM | WHAT GOES WRONG HERE |
|---|---|
| Priya moves up to Grade 10 | |
| Someone types ‘Priya Shaw’ on one row | |
| A new member joins but borrows nothing | |

Now split the loans table into two. Fill in the field names only — not the data.

| MEMBERS TABLE · FIELDS | LOANS TABLE · FIELDS |
|---|---|
| | |
| | |
| | |

Which field appears in **both** tables, and what is its job? ______________________________

# When keys go wrong

STRETCH

16 MIN

## Task 3.4 · Two records, one key

PREDICT

A careless import gives two different members the same MemberID: `M04`.

| # | QUESTION | YOUR ANSWER |
|---|---|---|
| a | What happens when the librarian searches for M04? | |
| b | What happens to a loan recorded against M04? | |
| c | Why can the database not simply pick the more likely one? | |

**d.** Write the rule, in plain English, that the database should have enforced.

## Task 3.5 · Write the query

EXTENSION

Write these two requests the way you would say them to a database: which table, which fields, which condition.

| PLAIN REQUEST | TABLE | FIELDS TO SHOW | CONDITION |
|---|---|---|---|
| Everyone in Grade 8 | | | |
| Every fiction book due after August 20 | | | |

**The last word.** The library could keep everything in one enormous table and it would still work. Give the strongest reason not to.

## Before you hand this in

CHECKPOINT

**Check what you managed today**

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

**One thing I now understand that I did not this morning**

Teacher initials ____________________

LESSON 03 · TEACHER COPY

# Answer key

Databases by Hand

NOT FOR STUDENTS

ANSWERS

## Task 3.1 · the key

**Full name** — no. Not unique; two students share it and the database cannot tell them apart.

**Email** — unique today, but it changes when a name changes and every old record points nowhere.

**Date of birth** — no. Nowhere near unique.

**Student ID** — **yes.** Unique, never empty, never reused, means nothing on its own.

**Phone number** — unique today, but people change numbers and numbers get reissued.

**The two are email and phone.** Both are unique but both *describe* the person rather than identify them, so both can change. A key must never change.

## Task 3.2 · the three queries

**a. Overdue today: L01, L04, L06, L09 — four.** Due date before August 12 and not returned. L03 and L08 are past due but were returned, which is the trap.

**b. Non-fiction: L02, L06, L07, L09 — four.**

**c. Priya (L01, L02), Ana (L04, L05, L09), Jed (L06, L10).** Tom has only L07 still out; L03 came back.

**d. Grade 9, not overdue: L02, L05 — two.**

**e.** Query c needs the rows *grouped* by member and counted. The other three test one row at a time; this one cannot be answered until every row has been seen.

## Task 3.3 · splitting the table

**MEMBERS**

MemberID (key) · Name · Grade

**LOANS**

LoanID (key) · MemberID · Title · Due · Returned

**MemberID appears in both.** In MEMBERS it is the primary key; in LOANS it is a **foreign key** — a pointer to a row in the other table. Accept ‘it links them’.

**The three problems:** Priya moving up means editing her year on every row she appears in, and missing one leaves the database disagreeing with itself · a misspelling splits one person into two and her loans stop adding up · a new member cannot be stored at all until she borrows something, because there is no row to put her in. Those three have names — update, inconsistency and insertion problems — but the reasoning is what earns the credit.

## Task 3.4 · the duplicate key

**a.** Two rows come back and the software has no way to choose. Some systems return the first and silently ignore the second, which is worse than an error.

**b.** The loan attaches to whichever record is found. One member is charged for a book the other borrowed.

**c.** Because both rows are equally valid. Nothing in the data says which is right — the information needed to decide was never stored.

**d.** ‘No two records in MEMBERS may share a MemberID, and MemberID may never be empty.’ That rule is what a primary key *is*.

## Task 3.5 · the queries and the last word

**Grade 8:** table MEMBERS · show Name · where Grade = 8.

**Fiction due after Aug 20:** table LOANS · show Title and Due · where Genre = Fiction AND Due > Aug 20. Strong students will notice Genre belongs in a BOOKS table, not LOANS — give full credit and say so.

**The last word:** one table works right up until something changes. Every fact stored twice is a chance for the two copies to disagree, and a database that disagrees with itself cannot be trusted for anything.

LESSON 04 · TEACHER BRIEFING

# Input, Process, Output

READ BEFORE THE BELL

2 MINUTES

Students break six everyday machines into input, process and output, sort ten components into sensors and actuators, then design the control rule for a school greenhouse and decide what it should do when a sensor fails.

### You do not need to know this subject

No electronics knowledge is needed. Every component in the lesson is named and described on the student page, and the key gives a model answer for each system.

### Photocopy these pages only

**Pages 22–24**

One per student. **Do not photocopy this page or the answer key.**

### What students need

A pen. Nothing else.

## WHAT TO SAY TO START

> *"Every machine in this building is doing the same three things: taking something in, deciding something, and making something happen. Once you can see those three, you can take apart anything."*

## HOW THE LESSON RUNS

| TIME | WHAT HAPPENS | WHAT YOU DO |
|---|---|---|
| 0–5 | Hand out sheets, settle | Use the door script on page 5 |
| 5–12 | Read the panel and the automatic door | Do the door on the board with them |
| 12–26 | Task 4.1 — six everyday systems | Several right answers. Key shows the pattern |
| 26–36 | Task 4.2 — sensor or actuator | Ten answers, all definite |
| 36–48 | Task 4.3 and 4.4 — design it, then break it | The failure question is the good one |
| 48–55 | Task 4.5, collect sheets | Discussion prompt below |

### Questions students will ask

**"Is a screen an input or an output?"**
An output. A touchscreen is both — the glass shows, the layer under it senses. Worth saying out loud.

**"Is a button a sensor?"**
Yes. It senses one thing: pressed or not pressed. Students find that surprising and it is a good argument to let run.

**"What if there are two outputs?"**
Most real systems have several. Ask them to list all of them rather than choosing.

### If they finish early

Task 4.4 asks what the greenhouse should do when a sensor breaks. Strong students can then work out which of the six systems in Task 4.1 would be dangerous if its sensor stuck, and which would merely be annoying.

### If the computers are working

Any block-based simulator with virtual sensors works, and so does a phone: the screen brightness sensor and the flashlight make a complete input-process-output loop students can watch.

### Curriculum focus · for the regular teacher's records

Digital systems and their components; input, process and output; sensors and actuators in controlled systems; designing a control algorithm. Discussion prompt for the last five minutes: *"A smoke alarm that has failed looks exactly like a smoke alarm that is working. How would you design one where that is not true?"*

LESSON 04 · INPUT, PROCESS, OUTPUT

# The three parts of every machine

UNDERSTAND

14 MIN

NAME ______

CLASS ______

DATE ______

### Input

Something the system measures or is told. A button, a sensor, a typed number.

### Process

The decision or calculation. Usually an IF. This is the only part you write.

### Output

Something that changes in the world. A motor, a light, a sound, a screen.

### Worked example · the automatic door

**Input:** a motion sensor above the door, reading movement or no movement.
**Process:** IF movement detected AND the door is closed, THEN open. Wait 4 seconds after the last movement, THEN close.
**Output:** the motor that drives the door.

Notice that the timer is part of the process, not the input. Nothing measures it — the system counts it.

## Task 4.1 · Take six machines apart

ANALYZE

| SYSTEM | INPUT | PROCESS | OUTPUT |
|---|---|---|---|
| Pedestrian crossing button | | | |
| Microwave oven | | | |
| Smoke alarm | | | |
| Parking lot gate | | | |
| Street light | | | |
| Automatic faucet in the restroom | | | |

One of these six has an input that is not a sensor at all. Which one? ______

# Sensing and doing

KEY SKILL

18 MIN

## Task 4.2 · Sensor or actuator?

CLASSIFY

Write **S** for sensor (it measures) or **A** for actuator (it does). Then say what it measures or does.

| COMPONENT | S/A | WHAT IT MEASURES OR DOES |
|---|---|---|
| Motion detector | | |
| Buzzer | | |
| Thermistor | | |
| Servo motor | | |
| Push button | | |

| COMPONENT | S/A | WHAT IT MEASURES OR DOES |
|---|---|---|
| Light sensor (LDR) | | |
| LED | | |
| Microphone | | |
| Heating element | | |
| Moisture probe | | |

Which one in this list could reasonably be argued into both columns, and why?

## Task 4.3 · Design the greenhouse

DESIGN

The school greenhouse must keep seedlings alive over the holidays. It has a water pump, a vent flap, a fan and a heat lamp available.

| WHAT MUST BE CONTROLLED | SENSOR YOU WOULD USE | THE RULE (IF … THEN …) |
|---|---|---|
| Soil moisture | | |
| Air temperature | | |
| Light after dark | | |

**a.** One rule must not run more than once a day even if the condition stays true. Which one, and why?

**b.** Two of your rules could fight each other. Which two, and what would you see?

LESSON 04 · INPUT, PROCESS, OUTPUT

# When the sensor lies

STRETCH

16 MIN

## Task 4.4 · The stuck sensor

PREDICT

A sensor that breaks rarely announces it. It usually just keeps reporting the last value it read, forever.

| # | THE FAILURE | WHAT THE SYSTEM DOES | WHO NOTICES, AND WHEN |
|---|---|---|---|
| a | Moisture probe stuck on ‘wet’ | | |
| b | Moisture probe stuck on ‘dry’ | | |
| c | Temperature sensor stuck on 72°F | | |

**d.** Write one rule the greenhouse could run that would catch a stuck sensor without any extra hardware.

## Task 4.5 · Fail safe or fail secure

THINK

When a system loses power, it has to choose. An electric door lock can fail **open** or fail **locked**.

| WHICH DOOR | SHOULD FAIL | WHY |
|---|---|---|
| A fire exit | | |
| A server room | | |

**The last word.** Explain in one sentence why a system that fails loudly is safer than one that fails quietly.

## Before you hand this in

CHECKPOINT

**Check what you managed today**

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

**One thing I now understand that I did not this morning**

Teacher initials ________________

# Answer key

Input, Process, Output

NOT FOR STUDENTS

ANSWERS

## Task 4.1 · model answers

| SYSTEM | INPUT | PROCESS | OUTPUT |
|---|---|---|---|
| Crossing button | Button press; traffic light timer | IF pressed, wait for the safe gap, then change | Red/green figure, beeper |
| Microwave | Keypad time and power; door switch | Count down; refuse to run if the door is open | Magnetron, turntable, beep |
| Smoke alarm | Particles in the chamber | IF above threshold for a few seconds, sound | Siren, flashing light |
| Parking gate | Ticket or plate reader; loop in the road | IF valid AND car present, raise | Gate motor, display |
| Street light | Light sensor, or a clock | IF darker than threshold, switch on | Lamp |
| Automatic faucet | Infrared reflection sensor | IF hand present, run for up to 15 seconds | Solenoid valve |

**The crossing button** is the odd one out: a person deliberately telling the system something, rather than the system measuring the world. Accept the microwave keypad as well — it is the same argument and worth crediting.

## Task 4.2 · sensor or actuator

**S** motion detector (movement) · **A** buzzer (makes sound) · **S** thermistor (temperature) · **A** servo (turns to an angle) · **S** button (pressed or not) · **S** LDR (light level) · **A** LED (light) · **S** microphone (sound) · **A** heating element (heat) · **S** moisture probe (water in soil).

**The arguable one is the LED.** Run backwards it generates a tiny current when light falls on it, and some cheap devices use one as both. Give full credit for any student who argues the buzzer both makes and detects vibration.

## Task 4.3 · the greenhouse

**Moisture:** probe. IF soil is dry THEN run the pump for 20 seconds.

**Temperature:** thermistor. IF above 85°F THEN open the vent and run the fan; IF below 50°F THEN close the vent and switch on the heat lamp.

**Light:** LDR or a clock. IF dark AND it is before 9pm THEN switch on the lamp.

**a.** The watering rule. Soil stays dry for minutes after watering, so a rule that keeps checking will pump again and again and drown the seedlings. It needs a wait, or one run per day.

**b.** The heat lamp and the vent. On a cold bright day the lamp heats the air, the vent opens to cool it, the temperature drops, the lamp runs harder. The system oscillates and wastes power.

## Task 4.4 · the stuck sensor

**a.** Stuck on wet: the pump never runs and the plants die of thirst. Nobody notices until the plants are visibly dead — weeks.

**b.** Stuck on dry: the pump runs constantly, the greenhouse floods, and this one is noticed fast because water appears where it should not.

**c.** Stuck on 72°F: vent and lamp never activate. The greenhouse cooks in summer and freezes in winter, and the log looks perfectly healthy the whole time.

**d.** Model answer: if a reading has not changed *at all* for 24 hours, raise an alert. Real sensors always wobble slightly; a perfectly constant value is the signature of a dead one. Accept 'check the pump ran but the moisture never changed' — that is even better, because it tests the whole loop.

## Task 4.5 · fail safe, fail secure

**Fire exit: fail open.** Being locked in a burning building is a worse outcome than a door that anyone can push.

**Server room: fail locked.** A power cut is exactly when an intruder would like the door to swing open.

The point: there is no universally safe default. The right answer depends entirely on what you are protecting and from what.

**The last word:** a system that fails quietly keeps producing plausible output that nobody questions, so the failure is discovered by its consequences instead of by its alarm. Accept any answer reaching 'you cannot fix a fault you do not know about'.

LESSON 05 · TEACHER BRIEFING

# Making It Smaller

READ BEFORE THE BELL

2 MINUTES

Students compress a picture by hand using run-length encoding, decode a second picture from its codes, then work out why the same method makes a photograph bigger. Puzzle-shaped and very markable.

### You do not need to know this subject

Every answer is a list of numbers that can be counted straight off the grid, and every grid is drawn out in the key.

### Photocopy these pages only

**Pages 27–29**

One per student. **Do not photocopy this page or the answer key.**

### What students need

A pen. A pencil is better for Task 5.2 because students shade squares.

## WHAT TO SAY TO START

> *“A photo on your phone is about four million pixels. The file is nowhere near four million numbers. Today you find out what happened to the rest of them.”*

## HOW THE LESSON RUNS

| TIME | WHAT HAPPENS | WHAT YOU DO |
|---|---|---|
| 0–5 | Hand out sheets, settle | Use the door script on page 5 |
| 5–14 | Read the panel and the worked row | Do row 2 on the board with them |
| 14–28 | Task 5.1 — encode the image | Exact answers. Grade as you circulate |
| 28–40 | Task 5.2 — decode the hidden image | It is a face. Let them find it |
| 40–52 | Tasks 5.3 and 5.4 — lossy, and the failure case | Key has model answers |
| 52–60 | Task 5.5, collect sheets | Discussion prompt below |

### Questions students will ask

**“Why start with white?”**
Because both ends have to agree on something. The rule here is that the first number is always the count of white pixels, even if it is zero.

**“My row adds up to 17, not 16.”**
Then one run is counted wrong. Every row must total exactly 16 — that check catches almost every mistake.

**“Is this how JPEG works?”**
No. This is the simplest real method and it is genuinely used in fax machines and simple graphics. JPEG throws information away, which is Task 5.3.

### If they finish early

Task 5.4 asks students to design the image that defeats the method. Strong students can then work out the exact size of the worst possible 16×8 image and explain why compression software checks whether it helped before saving.

### If the computers are working

Students can save the same photo as PNG and as JPEG at low quality and compare both file size and appearance. Zoom in to 400% on the JPEG and the discarded detail becomes visible.

### Curriculum focus · for the regular teacher’s records

How digital systems represent images; data compression; lossy and lossless techniques and their trade-offs. Discussion prompt for the last five minutes: *“A photo you have re-saved twenty times looks worse every time. A text file you have saved twenty thousand times does not. Why?”*

LESSON 05 · MAKING IT SMALLER

# Counting instead of listing

UNDERSTAND

16 MIN

| NAME | CLASS | DATE |
|---|---|---|
| | | |

## Run-length encoding

A black and white image is a grid of pixels. Storing it the obvious way means writing one number for every single pixel.

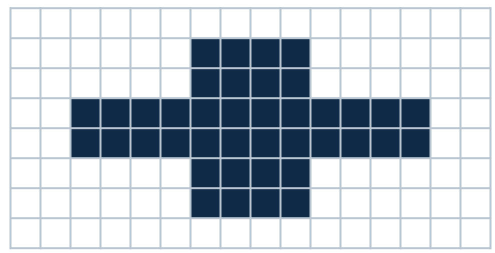
Instead, count the **runs**: how many white, then how many black, then how many white, and so on. The first number is always the count of white, even if it is zero. Nothing is lost — the original can be rebuilt exactly.

### Worked example · one row of 16

Six white, four black, six white.

**Code: 6, 4, 6** — three numbers instead of sixteen.

### The image for Task 5.1



## Task 5.1 · Encode the image

CALCULATE

| ROW | CODE | CHECK: NUMBERS TOTAL 16? | HOW MANY NUMBERS? |
|---|---|---|---|
| 1 | | | |
| 2 | | | |
| 3 | | | |
| 4 | | | |
| 5 | | | |
| 6 | | | |
| 7 | | | |
| 8 | | | |

**a.** Pixels in the original: 16 × 8 = ______ **b.** Numbers in your code: ______ **c.** How many times smaller? ______

# Decoding, and choosing what to lose

KEY SKILL

20 MIN

## Task 5.2 · Decode the hidden image

PUZZLE

### The codes

Row 1: 16

Row 2: 3, 10, 3

Row 3: 4, 2, 4, 2, 4

Row 4: 16

Row 5: 3, 1, 8, 1, 3

Row 6: 4, 8, 4

Row 7: 16

Row 8: 16

Shade the black runs. Remember the first number is always white.

What is it? ____________________

## Task 5.3 · Which kind of smaller?

CLASSIFY

**Lossless** rebuilds the original exactly. **Lossy** throws information away permanently to save more space. Write L for lossless or Y for lossy, and say why it must be that one.

| # | THE FILE | L / Y | WHY IT MUST BE THAT ONE |
|---|---|---|---|
| a | A photo posted to a social app | | |
| b | A program you downloaded in a zip file | | |
| c | A song streamed to your phone | | |
| d | A hospital X-ray | | |
| e | A spreadsheet of exam scores | | |

What do the lossless ones have in common? ______________________________

LESSON 05 · MAKING IT SMALLER

# The image that defeats it

STRETCH

16 MIN

## Task 5.4 · When compression makes it bigger

CHALLENGE

**Row X**

**Row Y**

| # | QUESTION | ANSWER |
|---|---|---|
| a | Code for row X | |
| b | Code for row Y | |
| c | Which row compressed well? | |
| d | How many numbers does row X need? | |

**e.** A photograph of a face has almost no two identical neighboring pixels. Explain what run-length encoding would do to it.

**f.** Compression software checks the result before it saves. What is it checking for?

## Task 5.5 · The last word

THINK

Re-saving a photograph as a lossy file over and over makes it visibly worse. Re-saving a text file a thousand times does not change it at all. Explain the difference in one sentence.

## Before you hand this in

CHECKPOINT

**Check what you managed today**

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

**One thing I now understand that I did not this morning**

Teacher initials

LESSON 05 · TEACHER COPY

# Answer key

Making It Smaller

NOT FOR STUDENTS

ANSWERS

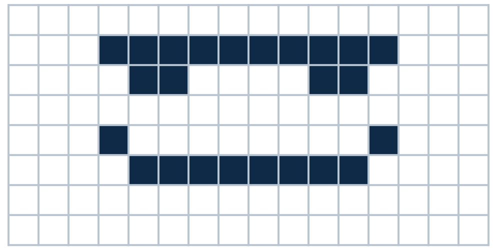
## Task 5.1 · the codes

Row 1: 16
Row 2: 6, 4, 6
Row 3: 6, 4, 6
Row 4: 2, 12, 2
Row 5: 2, 12, 2
Row 6: 6, 4, 6
Row 7: 6, 4, 6
Row 8: 16

**a.** 128 pixels. **b.** 20 numbers. **c.** 128 ÷ 20 = **6.4 times smaller**.

Every row must total 16. A student whose numbers total 15 or 17 has miscounted a run — send them back to the check column rather than grading it wrong.

## Task 5.2 · the hidden image



**It is a face** — a band across the top, two eyes, and a smile with turned-up corners. Accept ‘smiley’, ‘emoji’ or ‘face’.

Students who shade the first number black instead of white get a photographic negative of the face, which is a useful thing to point out: both ends of a compression must agree on the convention or the data is meaningless.

## Task 5.3 · lossy or lossless

| # | FILE | ANSWER | WHY |
|---|---|---|---|
| a | Photo on a social app | **Y** | Nobody can see the difference and the saving is huge |
| b | Program in a zip | **L** | Change one bit of a program and it will not run |
| c | Streamed song | **Y** | Sounds beyond hearing are discarded to fit the bandwidth |
| d | Hospital X-ray | **L** | The discarded detail could be the diagnosis |
| e | Spreadsheet of scores | **L** | An approximately correct score is a wrong score |

**The common thread:** lossless is required wherever every single value carries meaning. Lossy is acceptable wherever a human eye or ear is the only judge, because human senses miss things and the file can exploit that.

## Task 5.4 · the failure case

**a.** Row X: `0, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1` — sixteen numbers, plus the leading zero for ‘no white first’. Accept seventeen numbers with the zero, or sixteen without it, provided the reasoning is right.

**b.** Row Y: `0, 8, 8` .

**c.** Row Y — two runs instead of sixteen.

**d.** Sixteen or seventeen numbers to describe sixteen pixels. No saving at all, and worse than the original if each number needs more space than one pixel did.

**e.** It would make the file **bigger**. Every run is one pixel long, so the code is longer than the picture.

**f.** Whether the compressed version is actually smaller. If it is not, the software stores the original and marks it as uncompressed.

## Task 5.5 · the closing question

A lossy file is rebuilt as an approximation. Save it again and the next approximation is made from the approximation, not the original — so the errors accumulate, generation after generation.

Lossless compression rebuilds the file bit for bit, so the thousandth save is identical to the first. There is nothing to accumulate.

Accept any answer reaching ‘each save copies the copy, not the original’.

LESSON 06 · TEACHER BRIEFING

# Secret Writing

READ BEFORE THE BELL

2 MINUTES

Students encrypt and decrypt with a shift cipher, break one without being given the key, then calculate how long a modern key would take to guess. Puzzle-driven, entirely arithmetic-free until the last task.

### You do not need to know this subject

An alphabet strip is printed on the student page and every answer is a word or a number in the key. You do not need to be able to do any of it yourself.

### Photocopy these pages only

## Pages 32–34

One per student. **Do not photocopy this page or the answer key.**

### What students need

A pen. A calculator helps for Task 6.3 but the numbers are printed if they do not have one.

#### WHAT TO SAY TO START

> *"Writing a message nobody else can read is two and a half thousand years old. Today you will do it, then you will break someone else's, and then you will find out why the method you just used is now useless."*

#### HOW THE LESSON RUNS

| TIME | WHAT HAPPENS | WHAT YOU DO |
|---|---|---|
| 0–5 | Hand out sheets, settle | Use the door script on page 5 |
| 5–12 | Read the panel and the worked shift | Do one letter each way on the board |
| 12–26 | Task 6.1 — encrypt and decrypt | Self-checking: it reads or it does not |
| 26–40 | Task 6.2 — break it without the key | Hint is on this page if needed |
| 40–52 | Task 6.3 and 6.4 — keyspace and the key problem | Key has every number |
| 52–60 | Task 6.5, collect sheets | Discussion prompt below |

### Questions students will ask

**"What happens at Z?"**
It wraps back to A. Z shifted by 3 is C. The alphabet strip on the sheet makes this visible.

**"I cannot find the shift."**
Hint: the two-letter word in the message is almost certainly IS, IN, IT or AT. Try one and see if the rest falls out.

**"Can I just try all of them?"**
Yes, and that is exactly the point of Task 6.2. Twenty-five attempts is nothing. Twenty-five is why this cipher is dead.

### If they finish early

Task 6.4 asks how you send the key to someone you have never met. Strong students can then explain why writing the key on the same sheet as the message defeats the entire exercise.

### If the computers are working

Any online Caesar cipher tool will do the shifting instantly, which is a useful demonstration of Task 6.2 rather than a shortcut. Insist the paper work is finished first.

### Curriculum focus · for the regular teacher's records

Data security; encryption and decryption; the role of keys; evaluating the strength of a security measure. Discussion prompt for the last five minutes: *"Encryption protects the message. So what protects the key — and why is that the hard part?"*

LESSON 06 · SECRET WRITING

# Shifting the alphabet

UNDERSTAND

16 MIN

NAME

CLASS

DATE

## The oldest cipher there is

A **shift cipher** moves every letter the same number of places along the alphabet. The number is the **key**. Both people must know it and nobody else may.

With a key of 3: A becomes D, B becomes E, and Z wraps around to C. To decrypt, shift back.

## The alphabet strip

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

Tear this strip out mentally and slide it along. Wrap Z round to A.

## Task 6.1 · Encrypt and decrypt

DO

**a.** Encrypt with a key of **5** (shift forwards):

| PLAIN | ENCRYPTED |
|---|---|
| MEET AT NOON | |
| SEND HELP | |

**b.** Decrypt with a key of **3** (shift backwards):

| ENCRYPTED | PLAIN |
|---|---|
| WKH GRRU LV RSHQ | |
| ORFN LW | |

**c.** If the key is 26, what does the message look like? Why? ____________________

# Breaking it without the key

KEY SKILL
18 MIN

## Task 6.2 · You were not given the key

CHALLENGE

This message was intercepted. Nobody told you the shift.

intercepted CIPHERTEXT

AOL WHZZDVYK PZ VWLU

| # | QUESTION | ANSWER |
|---|---|---|
| a | How many keys are there to try? | |
| b | Which word did you attack first, and why? | |
| c | The key is | |
| d | The message says | |

**e.** Roughly how long would twenty-five attempts take you by hand? And a computer?

______________________________

## Task 6.3 · How big is the haystack?

CALCULATE

A computer that tries **one billion** keys every second is attacking each of these. Fill in the last column: seconds, hours, years or longer.

| THE SECRET | POSSIBLE VALUES | TIME TO TRY THEM ALL |
|---|---|---|
| A shift cipher key | 25 | |
| A 4-digit PIN | 10,000 | |
| 6 lower-case letters | 308,915,776 | |
| 8 mixed characters | $6 \times 10^{15}$ | |
| A modern 128-bit key | $3 \times 10^{38}$ | |

Each extra character multiplies the work. Explain why that is better than making the algorithm more complicated. ______________________________

# The problem nobody mentions

STRETCH

16 MIN

## Task 6.4 · How do you send the key?

THINK

You want to send an encrypted message to someone in another country. You have never met them and every channel you have can be read by somebody else.

| # | QUESTION | YOUR ANSWER |
|---|---|---|
| a | Why can you not simply email them the key? | |
| b | Why does writing the key on the message defeat the whole exercise? | |
| c | You share one key with fifty people. One of them leaks it. What now? | |

**d.** Suggest one way two strangers could agree on a secret while being overheard the whole time. There is a real answer, and guessing at it is the point.

## Task 6.5 · The last word

THINK

Every cipher in this lesson is unbreakable if the key stays secret and useless the moment it does not. Explain in one sentence what that means for where the real weakness in any secure system usually sits.

## Before you hand this in

CHECKPOINT

**Check what you managed today**

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

**One thing I now understand that I did not this morning**

Teacher initials

LESSON 06 · TEACHER COPY

# Answer key

Secret Writing

NOT FOR STUDENTS

ANSWERS

## Task 6.1 · encrypting and decrypting

**a.** MEET AT NOON → **RJJY FY STTS**
SEND HELP → **XJSI MJQU**

**b.** WKH GRRU LV RSHQ → **THE DOOR IS OPEN**
ORFN LW → **LOCK IT**

**c.** A key of 26 shifts every letter all the way round to itself, so the message is sent in plain text. There are only **25 useful keys**, not 26 — a nice observation and worth a mark.

## Task 6.2 · breaking it

**a.** 25.

**b.** PZ — a two-letter word in English is a very short list, and IS fits immediately.

**c.** The key is **7**.

**d. THE PASSWORD IS OPEN**.

**e.** By hand, perhaps ten minutes. A computer tries all 25 in far less than a thousandth of a second. That gap is the entire reason this cipher belongs in a museum.

## Task 6.3 · the keyspace table

| THE SECRET | VALUES | AT A BILLION GUESSES A SECOND |
|---|---|---|
| Shift cipher | 25 | Instant — far under a millionth of a second |
| 4-digit PIN | 10,000 | Instant. This is why a phone limits attempts instead |
| 6 lower-case letters | 308,915,776 | About a third of a second |
| 8 mixed characters | $6 \times 10^{15}$ | About 70 days |
| 128-bit key | $3 \times 10^{38}$ | Longer than the age of the universe, by a very long way |

**Why length beats cleverness:** a longer key multiplies the attacker's work every single time you add a character, and it costs the defender nothing. A cleverer algorithm has to be kept secret to help, and secrets leak. Serious cryptography publishes the algorithm and protects only the key — worth saying out loud to a class that assumes the opposite.

The PIN row is the one to dwell on. A four-digit PIN is trivially guessable, and it is safe on a phone only because the phone refuses after a few tries. The protection is the lock-out, not the number.

## Task 6.4 · the key exchange problem

**a.** Anyone reading the message can read the key, so the encryption protects nothing.

**b.** Same reason, more obviously: you have posted the lock and the key together.

**c.** Everything ever sent with that key is now readable, including messages sent before the leak, and all fifty people need a new key delivered somehow. That is the reason real systems give every pair of people a different key.

**d.** There is a genuine answer and no student is expected to reach it: the two strangers exchange *public* keys that anyone may see, and each keeps a private key that never travels. Anything locked with the public key can only be opened with the private one. Credit any answer that reaches 'send something that is useless to an eavesdropper but useful to the recipient'.

## Task 6.5 · the closing question

The mathematics is almost never the weak point. The weak point is wherever a human holds, moves, reuses or writes down the key.

Accept any answer reaching 'the people, not the math'. If the class has already done Volume 1 Lesson 05 on how attacks actually work, this is the same conclusion arriving from the opposite direction, and it is worth pointing that out.

LESSON 07 · TEACHER BRIEFING

# Design for a Real User

READ BEFORE THE BELL

2 MINUTES

Students turn a vague brief into testable design criteria, sketch an interface for a real user with real constraints, and diagnose six interfaces that were designed for the wrong person.

### You do not need to know this subject

No technology and no drawing skill is needed. Boxes and labels are enough, and the key grades structure rather than neatness.

### Photocopy these pages only

## Pages 37–39

One per student. **Do not photocopy this page or the answer key.**

### What students need

A pen. A ruler is useful for the sketch but not required.

#### WHAT TO SAY TO START

> *“The person who designed the school printer knew exactly how it worked. That is the problem. Today you design for somebody who is not you.”*

#### HOW THE LESSON RUNS

| TIME | WHAT HAPPENS | WHAT YOU DO |
|---|---|---|
| 0–5 | Hand out sheets, settle | Use the door script on page 5 |
| 5–12 | Read the brief together | Have a student read it aloud |
| 12–26 | Task 7.1 — turn the brief into criteria | Testable is the word to keep repeating |
| 26–40 | Task 7.2 — sketch the screen | Quiet working time. Labels matter, art does not |
| 40–50 | Task 7.3 — six broken interfaces | Six answers, all in the key |
| 50–55 | Tasks 7.4 and 7.5, collect sheets | Discussion prompt below |

### Questions students will ask

**“I cannot draw.”**
Rectangles and words. Say this before they start or a third of the class will not begin.

**“How is my criterion wrong? It says it should be easy.”**
Ask how you would prove it. If two people can disagree about whether it was met, it is not a criterion yet.

**“Can I add features?”**
Only if the brief asks for them. Adding is the easy part; the brief has constraints for a reason.

### If they finish early

Task 7.4 asks students to write a test plan for a real user. Strong students should then be pushed to say what they would do if the test user failed — because the instinct to explain is exactly the instinct to resist.

### If the computers are working

Any drawing or slides tool works for a wireframe, and most students find it slower than paper. The better on-screen task is to find three real apps that solve the same problem and compare their first screens.

### Curriculum focus · for the regular teacher’s records

Defining problems and design criteria; user-centered design; generating and evaluating designed solutions; usability and accessibility.
Discussion prompt for the last five minutes: *“You designed it, so you know where everything is. Explain why that makes you the worst possible person to test it.”*

# Reading a brief properly

UNDERSTAND

14 MIN

NAME

CLASS

DATE

## The brief

The school office wants a screen that lets a student sign in when they arrive late. At the moment there is a paper book and a line of eleven people at 8:55am.

**Constraints:** one shared tablet on a stand in the foyer · the whole interaction must take under fifteen seconds · students will not be logged in · the office must be able to read the day's list without training · nothing may be installed on student phones.

## Task 7.1 · Turn it into criteria

PROVE

A design criterion is a statement you can **test**. "It should be easy to use" is not one, because two people can disagree about whether it was met.

| VAGUE VERSION | WRITTEN AS A TESTABLE CRITERION |
|---|---|
| It should be quick | |
| It should be easy to use | |
| It should work for everyone | |
| It should look good | |
| The office should be able to use it | |

**a.** Who is the user? There is more than one. ______________________

**b.** Which constraint would be hardest to design around, and why? ______________________

# Draw the screen

KEY SKILL

20 MIN

## Task 7.2 · The sign-in screen

DRAW

Sketch the **first** screen a late student sees. Boxes and labels only. Include and label all six of these:

- Where the student identifies themselves
- How they say why they are late
- The confirmation that it worked
- A way to correct a mistake
- What happens when two students arrive together
- What the screen shows when nobody is using it

## Task 7.3 · Designed for the wrong person

DIAGNOSE

| # | WHAT THE INTERFACE DOES | WHO DOES THIS FAIL, AND WHY |
|---|---|---|
| a | Errors are shown by turning the field red, with no message | |
| b | Delete sits directly beside Save, same size, same color | |
| c | The form times out after 30 seconds and clears itself | |
| d | The only way to continue is to double-click a small icon | |
| e | The button says `COMMIT TRANSACTION` | |
| f | There is no way to undo anything | |

Two of these six would fail a person with perfectly good eyesight and no disability at all. Which two?

____________________

LESSON 07 · DESIGN FOR A REAL USER

# Test it on somebody else

STRETCH

16 MIN

## Task 7.4 · The test plan

EXTENSION

You are going to put your sketch in front of one person who has never seen it. Write three tasks you would ask them to attempt, and what you would record.

| # | ASK THEM TO… | WHAT I WOULD WRITE DOWN |
|---|---|---|
| 1 | | |
| 2 | | |
| 3 | | |

**a.** They get stuck. What is the one thing you must not do? ____________________

**b.** They finish everything perfectly first time. What have you actually learned?

____________________

## Task 7.5 · The last word

THINK

“It is not confusing, they just did not read it properly.” Explain in one sentence why a designer who says this has stopped designing.

____________________

____________________

____________________

## Before you hand this in

CHECKPOINT

**Check what you managed today**

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

**One thing I now understand that I did not this morning**

____________________

____________________

Teacher initials ____________________

# Answer key

Design for a Real User

NOT FOR STUDENTS

ANSWERS

## Task 7.1 · model criteria

| VAGUE | TESTABLE |
|---|---|
| Quick | A student can complete sign-in in under 15 seconds without help |
| Easy to use | A student who has never seen it finishes it first time with no instructions |
| Works for everyone | Usable one-handed, readable from 3 feet, no color-only information |
| Looks good | Text is at least 18pt; no more than three colors; matches school branding |
| Office can use it | The office can produce today’s list in under three clicks |

**a. There are at least three users:** the late student, the office staff who read the list, and arguably the next student in the queue. Most classes name only the first. Push for the second — it is the one real projects forget.

**b.** Usually ‘students will not be logged in’: without a login the system has no idea who is standing there, so identification has to be fast, public and hard to fake all at once. Accept the fifteen seconds with good reasoning.

## Task 7.2 · grading the sketch

Credit is for structure, not neatness. Look for:

- All six required elements present and labeled
- Identification that does not require typing a full name
- A confirmation that is visible from standing distance
- A visible way back or a cancel
- An idle state that shows nothing about the previous student
- One clear action per screen

**The best answers handle two students arriving together** by returning to the idle screen immediately after a confirmation, rather than leaving the last student’s name on display. That is a privacy decision hiding inside a design decision, and it is worth pointing out.

## Task 7.3 · the six failures

**a.** Anyone who is color-blind — and anyone who now knows something is wrong but not what.

**b.** Everybody. Destructive and safe actions must never look alike or sit together.

**c.** Anyone slower: a person with a motor impairment, a person using a second language, a person holding a bag.

**d.** Anyone with limited fine motor control, and anyone on a touchscreen — where double-click is not really a gesture at all.

**e.** Everyone who is not a programmer. The label describes the code, not the user’s intention.

**f.** Everyone, eventually. Every person makes mistakes; only some interfaces forgive them.

**The two that fail anybody at all: b and f.** Accept e as well with reasoning.

## Task 7.4 · testing

Good tasks are goals, not instructions: ‘you arrived at 9:10 because the bus was late — sign in’, not ‘tap the button in the corner’.

Good things to record: where they hesitated, what they touched first, what they said out loud, how long it took, and where they were wrong about what would happen next.

**a.** You must not help. The moment you explain, you have destroyed the only data the test was going to produce — and in real use you will not be standing there.

**b.** Very little. One user succeeding proves the design is not catastrophically broken; it does not prove it works. That answer is worth credit on its own.

## Task 7.5 · the closing question

If users consistently do not read it, that is a finding about the design, not a fault in the users. Blaming the user converts every piece of evidence into an excuse, and a designer who does that can no longer learn anything from testing.

Accept any answer reaching ‘the user is not wrong, the design is’.

LESSON 08 · TEACHER BRIEFING

# Break It Into Blocks

READ BEFORE THE BELL

2 MINUTES

Students find the repetition in a long routine, name it, work out what goes in and what comes back, then meet the moment a variable inside a function refuses to exist outside it. The hardest lesson here.

### You do not need to know this subject

Every program is under ten lines and the plain-English purpose is written beside each one. The key gives the answer and the reason for every part.

### Photocopy these pages only

**Pages 42–44**

One per student. **Do not photocopy this page or the answer key.**

### What students need

A pen. Pairs work well for this lesson.

## WHAT TO SAY TO START

> *“Professional programs are not long. They are short programs stacked on top of each other, each one with a name. Today you find out how that stacking works, and why it is the only way anyone copes.”*

## HOW THE LESSON RUNS

| TIME | WHAT HAPPENS | WHAT YOU DO |
|---|---|---|
| 0–5 | Hand out sheets, settle | Use the door script on page 5 |
| 5–14 | Read the worked example together | Do the three calls on the board |
| 14–28 | Task 8.1 and 8.2 — spot it, name it | Circulate. Key has model names |
| 28–42 | Task 8.3 — trace the call | The last line is the whole lesson |
| 42–54 | Task 8.4 — decompose a real job | Pairs are fine here |
| 54–60 | Task 8.5, collect sheets | Discussion prompt below |

### Questions students will ask

**“Do I need to know Python?”**
No. Every line has a plain-English comment. Say this out loud before they start.

**“Why does the last line crash?”**
That is Task 8.3d and it is the point of the lesson. Do not give it away — let them predict first.

**“What is the difference between a parameter and a variable?”**
A parameter is a variable that gets its value from whoever calls the function. Everything else about it is the same.

### If they finish early

Task 8.4 asks students to decompose a real job into named blocks. Strong students can then identify which of their blocks could be reused in a completely different project, which is the actual reason professionals write them.

### If the computers are working

Students can type the Task 8.3 program in and run it. The error message on the final line is far more convincing than any explanation — ask them to predict the message before running.

### Curriculum focus · for the regular teacher’s records

Decomposing problems into modules; defining and calling functions with parameters and return values; variable scope; designing for reuse. Discussion prompt for the last five minutes: *“A 300-line program with no functions and a 400-line program built from twelve of them. The second is longer. Why is it easier to fix?”*

LESSON 08 · BREAK IT INTO BLOCKS

# Naming a chunk of work

UNDERSTAND

16 MIN

NAME ____________ CLASS ____________ DATE ____________

## A function is a named box of instructions

You put something **in** (the parameters), it does its work, and it hands something **back** (the return value). Once it exists you never write those steps again — you just say its name.

Everything inside the box is private. The rest of the program cannot see it, and does not need to.

worked.py — PYTHON

```
def area(w, h):
    return w * h

print(area(3, 4))
print(area(10, 2))
print(area(7, 7))
```

### What happened

The multiplication is written **once** and used three times.

**Parameters:** w and h. **Returns:** one number.

Output: `12`, `20`, `49`.

To change how area is calculated you now change one line, not three.

## Task 8.1 · Find the repetition

DETECTIVE

This routine prints a report card for three students. Circle every group of lines that does the same job as another group.

report.py — PYTHON

```
print("NAME: Priya")
total = 82 + 74 + 90
print("AVG:", total / 3)
if total / 3 >= 50: print("PASS")

print("NAME: Tom")
total = 41 + 55 + 48
print("AVG:", total / 3)
if total / 3 >= 50: print("PASS")

print("NAME: Ana")
total = 66 + 71 + 59
print("AVG:", total / 3)
if total / 3 >= 50: print("PASS")
```

**a.** How many times is the same work written? ______ **b.** What changes between the copies?

____________________

**c.** Name the function you would write: `def ____________(____________)`

LESSON 08 · BREAK IT INTO BLOCKS

# In, out and invisible

KEY SKILL

20 MIN

## Task 8.2 · What goes in, what comes back

PROVE

| FUNCTION | WHAT IT SHOULD DO | PARAMETERS | WHAT IT RETURNS |
|---|---|---|---|
| is_weekend | Says whether a day is Saturday or Sunday | | |
| average | Works out the mean of a list of scores | | |
| ticket_price | Gives the fare for a passenger of a given age | | |
| shuffle | Puts a list of names into random order | | |
| print_banner | Prints a title with a line under it | | |

One of these five returns nothing at all. Which one, and is that a mistake?

______________________________

## Task 8.3 · Trace the call

TRACE

scope.py — PYTHON

```python
def double(n):
    result = n * 2
    return result

x = 5
y = double(x)
print(x, y)
print(result)
```

| # | QUESTION | ANSWER |
|---|---|---|
| a | What is n when the function runs? | |
| b | What does line 7 print? | |
| c | Does x change? Why? | |
| d | What happens on line 8? | |

**e.** Explain why the answer to d is a *useful* feature and not an annoying rule.

LESSON 08 · BREAK IT INTO BLOCKS

# Decompose something real

STRETCH

16 MIN

## Task 8.4 · Five named blocks

EXTENSION

Take the job of **running the school track and field day**. Break it into five named blocks. For each, say what it needs handed to it and what it hands back.

| BLOCK NAME | NEEDS (IN) | PRODUCES (OUT) |
|---|---|---|
| | | |
| | | |
| | | |
| | | |
| | | |

**a.** Which one block could be used unchanged for the swim meet? ______________________

**b.** Which one is hardest to describe in terms of in and out, and why? ______________________

## Task 8.5 · The last word

THINK

A 300-line program with no functions, and a 400-line program built from twelve. The second is longer. Explain in one sentence why it is easier to fix.

______________________

______________________

______________________

## Before you hand this in

CHECKPOINT

**Check what you managed today**

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

**One thing I now understand that I did not this morning**

______________________

______________________

Teacher initials ______________________

LESSON 08 · TEACHER COPY

# Answer key

Break It Into Blocks

NOT FOR STUDENTS

ANSWERS

## Task 8.1 · the repetition

**a.** Three times — four lines, repeated identically in structure.

**b.** Only the name and the three scores change. Everything else is a copy.

**c.** Model answer: `def report(name, m1, m2, m3)`. Accept `def report(name, scores)` — passing the list is better and worth saying so.

The routine drops from fifteen lines to a four-line function plus three calls. More importantly, changing the pass mark now means editing one line rather than three, and forgetting one of three is exactly how real bugs are born.

## Task 8.2 · in and out

| FUNCTION | IN | OUT |
|---|---|---|
| `is_weekend` | A day name | True or False |
| `average` | A list of numbers | One number |
| `ticket_price` | An age | A price |
| `shuffle` | A list | A list |
| `print_banner` | A title | Nothing |

**`print_banner` returns nothing, and that is correct.** Its whole job is the side effect — putting text on the screen. Functions come in two kinds: those that work something out and hand it back, and those that make something happen. Confusing the two is a common source of bugs.

## Task 8.3 · the trace

- **a.** n is 5 — the value of x was copied in.
- **b.** `5 10`.
- **c.** No. The function was handed a copy of the value, not the box itself.
- **d.** It crashes. `NameError: name 'result' is not defined`. `result` existed only while the function was running.
- **e.** Because every function can use short obvious names without any risk of clashing with a name somewhere else in a program you did not write.

This is **scope**, and it is the idea the whole lesson exists for. Without it, a program of any size would be a single shared pool of names in which every new line risks overwriting something a hundred lines away. Students find the crash irritating; it is the feature protecting them.

## Task 8.4 · decomposing the field day

A strong answer looks like: `register_students(grade levels → entry list)` · `build_program(entry list → event schedule)` · `assign_officials(staff list, schedule → duty schedule)` · `record_result(event, place, time → nothing)` · `calculate_house_points(results → house totals)`.

**a.** `calculate_house_points` and `assign_officials` transfer to the swim meet untouched — they never mention running. That is reuse, and it is what naming things buys you.

**b.** Usually `record_result`: it returns nothing and changes stored data instead. Same distinction as `print_banner` in Task 8.2, arriving from a completely different direction.

## Task 8.5 · the closing question

In the longer program every idea has a name and lives in one place, so a fault can be located, tested and repaired without reading the rest. In the shorter one, any line might depend on any other line, so understanding a bug means holding all 300 lines in your head at once.

Accept any answer reaching ‘you only have to understand one block at a time’. The best answers add that you can test each block on its own, which is genuinely the professional argument.

LESSON 09 · TEACHER BRIEFING

# Planning a Project

READ BEFORE THE BELL

2 MINUTES

Students break a real project into tasks, build a Gantt chart on paper, find the chain that decides the finish date, and discover which tasks can slip without costing a single day.

### You do not need to know this subject

Everything is addition and shading squares. The completed chart and the finish date are in the key.

### Photocopy these pages only

## Pages 47–49

One per student. **Do not photocopy this page or the answer key.**

### What students need

A pen. A ruler is genuinely useful here.

## WHAT TO SAY TO START

> *"Every group project you have ever done ran out of time in the last two days. Today you find out why that happens, and it is not because anybody was lazy."*

## HOW THE LESSON RUNS

| TIME | WHAT HAPPENS | WHAT YOU DO |
|---|---|---|
| 0–5 | Hand out sheets, settle | Use the door script on page 5 |
| 5–12 | Read the task list and the dependency idea | The word is: what must finish first |
| 12–24 | Task 9.1 — order the tasks | Several orders work. Key shows the constraint |
| 24–40 | Task 9.2 — shade the chart | Quiet working. Circulate with the key |
| 40–50 | Task 9.3 — what if it rains | The slack question is the good one |
| 50–55 | Tasks 9.4 and 9.5, collect sheets | Discussion prompt below |

### Questions students will ask

**"Can two tasks happen on the same day?"**
Yes, if different people do them and neither is waiting on the other. That is the entire point of the chart.

**"Does day 1 mean Monday?"**
Treat them as ten school days in a row. Weekends are ignored to keep the arithmetic clean.

**"My finish day is different from my partner's."**
One of you started a task before the thing it depends on had finished. That is the most common error and it is worth finding together.

### If they finish early

Task 9.4 is a risk register. Strong students can then work out what happens to the finish date if the task with the most slack is delayed by four days rather than one, and explain the difference.

### If the computers are working

Any spreadsheet makes a serviceable Gantt chart by shading cells, and project tools will draw the dependency arrows automatically. Ask students to predict the finish date before the tool tells them.

### Curriculum focus · for the regular teacher's records

Planning and managing digital projects; task breakdown and sequencing; identifying dependencies, milestones and risks. Discussion prompt for the last five minutes: *"A plan is wrong the moment something unexpected happens, which is always. So why write one?"*

LESSON 09 · PLANNING A PROJECT

# Tasks, order and dependencies

UNDERSTAND

14 MIN

| NAME | CLASS | DATE |
| --- | --- | --- |
| | | |

## The project

Your class has ten school days to produce a 90-second promotional video for the school open night. Six tasks have to happen. Some of them cannot start until others have finished.

| TASK | WHAT IT IS | DAYS | CANNOT START UNTIL |
| --- | --- | --- | --- |
| A | Write the script | 2 | — |
| B | Collect permission forms | 1 | — |
| C | Film the footage | 3 | A and B are finished |
| D | Edit the video | 3 | C is finished |
| E | Choose the music | 1 | — |
| F | Show a teacher and fix the notes | 1 | D is finished |

## Task 9.1 · Work out the order

PLAN

| # | QUESTION | ANSWER |
| --- | --- | --- |
| a | Which tasks could all start on day 1? | |
| b | What is the earliest day C could start? | |
| c | Which task decides when C can start — A or B? Why? | |
| d | How many days of work are there in total, added up? | |
| e | Will the project take that many days? Why not? | |

LESSON 09 · PLANNING A PROJECT

# Build the chart

KEY SKILL

20 MIN

## Task 9.2 · Shade the schedule

DRAW

Shade one box per day of work, starting each task on the earliest day it is allowed to start.

| TASK | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| A | | | | | | | | | | |
| B | | | | | | | | | | |
| C | | | | | | | | | | |
| D | | | | | | | | | | |
| E | | | | | | | | | | |
| F | | | | | | | | | | |

**a.** The video is finished at the end of day ______ **b.** Days spare before the deadline: ______

**c.** Write down the chain of tasks that decides the finish date: ______________________________

## Task 9.3 · Now it rains

PREDICT

| # | THE CHANGE | NEW FINISH DAY | WHY |
|---|---|---|---|
| a | Filming loses a day to rain (C takes 4) | | |
| b | Permission forms take 3 days instead of 1 | | |
| c | Choosing the music takes 4 days instead of 1 | | |
| d | Editing is done in 2 days instead of 3 | | |

Two of these four changed nothing at all. Those tasks are said to have **slack**. Which two, and roughly how much slack does each have? ______________________________

# What could go wrong

STRETCH

16 MIN

## Task 9.4 · The risk register

EXTENSION

For each risk, rate how likely it is (L/M/H), how bad it would be (L/M/H), and write one thing you would do **in advance** to reduce it.

| RISK | LIKELY | IMPACT | WHAT YOU WOULD DO ABOUT IT NOW |
|---|---|---|---|
| The main student in the video is away on filming day | | | |
| The footage is on one laptop and it fails | | | |
| The music you chose is copyrighted | | | |
| The teacher wants major changes at task F | | | |

Which of these four is best handled by changing the *plan* rather than by being careful?

______________________________

## Task 9.5 · The last word

THINK

Every plan is wrong within a week. Explain in one sentence why writing one is still worth the hour it takes.

______________________________

______________________________

______________________________

## Before you hand this in

CHECKPOINT

**Check what you managed today**

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

**One thing I now understand that I did not this morning**

______________________________

______________________________

Teacher initials ____________________

# Answer key

Planning a Project

NOT FOR STUDENTS

ANSWERS

## Task 9.1 · the order

**a.** A, B and E — none of them waits for anything.

**b.** Day 3. A needs days 1 and 2; B finishes on day 1.

**c.** **A**, because it takes longer. C waits for the *later* of the two to finish, not the earlier.

**d.** 2+1+3+3+1+1 = **11 days of work**.

**e.** No — it takes 9 days, because B and E happen alongside A instead of after it. Total effort and elapsed time are different quantities, and confusing them is why group projects overrun.

## Task 9.2 · the completed chart

| TASK | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| A | ■ | ■ | | | | | | | | |
| B | ■ | | | | | | | | | |
| C | | | ■ | ■ | ■ | | | | | |
| D | | | | | | ■ | ■ | ■ | | |
| E | ■ | | | | | | | | | |
| F | | | | | | | | | ■ | |

**a.** End of **day 9**. **b.** One day spare. **c.** The chain is **A → C → D → F** — 2+3+3+1 = 9. That chain has a name worth giving them: the **critical path**. Every day lost on it is a day lost on the project.

## Task 9.3 · what changes and what does not

| # | CHANGE | FINISH | WHY |
|---|---|---|---|
| a | C takes 4 days | **Day 10** | C is on the critical path. Every day added is a day lost |
| b | B takes 3 days | **Day 9** | B finishes on day 3, and C could not start before day 3 anyway |
| c | E takes 4 days | **Day 9** | Nothing waits for E until editing begins on day 6 |
| d | D takes 2 days | **Day 8** | Shortening a critical-path task pulls the whole finish forward |

**B and E have slack:** B can take up to 2 extra days and E up to 4 before either starts costing the project anything. This is the single most useful idea on the page — it tells you which tasks to worry about and, just as importantly, which ones to stop worrying about.

## Task 9.4 · the risk register

Model answers: **student away** — likely M, impact H; film their scenes first, or script a second presenter · **laptop fails** — likely L, impact H; copy the footage to a second place the same day · **copyrighted music** — likely H, impact M; choose from a licensed library before editing starts · **major changes at F** — likely M, impact H.

**The last one is the plan problem.** Being careful does not prevent it. Showing the teacher a rough cut on day 6 instead of a finished video on day 9 does — you move the feedback earlier, when changing things is still cheap. That is a change to the schedule, not to anyone's effort.

## Task 9.5 · the closing question

The value is not the schedule; it is knowing what depends on what. When something goes wrong, a plan tells you within a minute whether you have just lost a day or lost nothing at all — and which task to protect next.

Accept any answer reaching ‘it tells you what the delay costs’ or ‘it shows you what to fix first’.

LESSON 10 · TEACHER BRIEFING

# What a Free App Actually Costs

READ BEFORE THE BELL

2 MINUTES

Students work out how six free apps make money, audit a flashlight app that wants your contacts, and read five real terms-of-service clauses. Discussion-heavy and the easiest lesson here to supervise.

### You do not need to know this subject

No technology is involved and there is no single correct opinion. The key gives you the reasoning to look for and flags the three clauses that should worry a reader.

### Photocopy these pages only

**Pages 52–54**

One per student. **Do not photocopy this page or the answer key.**

### What students need

A pen. No devices needed, and do not ask students to open their own accounts.

## WHAT TO SAY TO START

> *"Nobody builds an app for forty million people out of kindness. Today you work out who is paying, what they are paying with, and whether it is you."*

## HOW THE LESSON RUNS

| TIME | WHAT HAPPENS | WHAT YOU DO |
|---|---|---|
| 0–5 | Hand out sheets, settle | Use the door script on page 5 |
| 5–12 | Read the four business models together | Ask which apps they use are free |
| 12–26 | Task 10.1 — match app to model | Expect disagreement. Ask for reasons |
| 26–38 | Task 10.2 — the permission audit | The flashlight app is the memorable one |
| 38–50 | Task 10.3 — read the terms | Three clauses should worry them |
| 50–55 | Tasks 10.4 and 10.5, collect sheets | Discussion prompt below |

### Questions students will ask

**"So is targeted advertising bad?"**
Not a settled question, and Task 10.5 exists to make them argue it rather than be told.

**"Everyone just clicks agree."**
Correct, and worth taking seriously. Ask whether that makes consent meaningless or simply practical.

**"What if I have nothing to hide?"**
Redirect: the question is not what you are hiding, it is who gets to decide what is known about you and for how long.

### If they finish early

Task 10.4 asks students to price the deal they would accept. Strong students can then argue the opposite case: what is genuinely lost if every useful app costs $4 a month up front?

### If the computers are working

Students can look at the permissions a real app on the school device requests, or read a published privacy policy. Keep it to apps nobody in the room has a personal account with.

### Curriculum focus · for the regular teacher's records

Impacts of digital systems on society; personal data, consent and privacy; economic and ethical considerations in the use of digital systems. Discussion prompt for the last five minutes: *"A targeted advertisement is more useful to you than a random one. Explain why that same sentence is both the strongest argument for and against the practice."*

LESSON 10 · WHAT A FREE APP ACTUALLY COSTS

# Somebody is paying

UNDERSTAND

14 MIN

NAME

CLASS

DATE

### Advertising

You are shown things. The more the app knows about you, the more an advertiser pays to reach you.

### Subscription

You pay money regularly. The app has no reason to collect anything it does not need.

### In-app purchase

The app is free; the good parts are not. Designed to make waiting unpleasant.

### Data itself

Your activity is combined with millions of others and sold or used to build something else.

## Task 10.1 · Who is paying, and with what?

ANALYZE

| # | THE APP | MODEL | WHAT IT GETS FROM YOU |
|---|---|---|---|
| a | A free game with $3 character outfits | | |
| b | A free map app run by a very large company | | |
| c | A weather app costing $6 once, no ads | | |
| d | A free social network with no visible ads yet | | |
| e | A free homework app paid for by an education department | | |
| f | A free flashlight app that wants your contacts | | |

One of these six is not a business at all. Which one, and who is the customer instead?

LESSON 10 · WHAT A FREE APP ACTUALLY COSTS

# Permissions and promises

KEY SKILL

20 MIN

## Task 10.2 · The flashlight app audit

JUDGE

A free flashlight app requests these permissions on install. For each, decide whether it is **needed**, **arguable** or **unjustifiable**, and say what it would allow.

| PERMISSION | VERDICT | WHAT IT WOULD ALLOW |
|---|---|---|
| Camera | | |
| Contacts | | |
| Precise location | | |
| Microphone | | |
| Run at startup | | |
| Full network access | | |

**a.** Which single permission would you refuse first, and what would the app most likely do about it?

______________________________

## Task 10.3 · Read the terms

INVESTIGATE

Five clauses from a fictional app's terms. Mark each **expected**, **unusual** or **unacceptable**, and write what it actually allows the company to do.

| # | THE CLAUSE | VERDICT | WHAT IT ALLOWS |
|---|---|---|---|
| 1 | You must be at least 13 to hold an account | | |
| 2 | We may use content you upload in our advertising, worldwide, without payment | | |
| 3 | We may share your information with partners and affiliates | | |
| 4 | We may change these terms at any time; continued use means acceptance | | |
| 5 | Deleted accounts are removed from view but retained on our systems | | |

LESSON 10 · WHAT A FREE APP ACTUALLY COSTS

# What would you actually accept?

STRETCH

16 MIN

## Task 10.4 · Write the fair deal

DECIDE

**a.** Write, in one sentence, the trade you would genuinely accept from a free app.

**b.** Same app, no advertising and no data collection. What is the most you would pay per month? ______

**c.** If every student in this room paid that, would it be enough to build the app? What does your answer suggest about why free won? ______

**d.** Name one thing you would refuse to trade at any price.

## Task 10.5 · The last word

THINK

A targeted advertisement is more useful to you than a random one. Explain in one or two sentences why that is both the best argument for the practice and the best argument against it.

## Before you hand this in

CHECKPOINT

**Check what you managed today**

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

**One thing I now understand that I did not this morning**

Teacher initials ______

# Answer key

What a Free App Actually Costs

NOT FOR STUDENTS

ANSWERS

## Task 10.1 · who pays

| # | APP | MODEL | WHAT IT TAKES |
|---|---|---|---|
| a | Free game, paid outfits | In-app purchase | Money from a small minority of players; attention from the rest |
| b | Free map app | Data and advertising | Where you go, when, and how often — which also improves the map |
| c | $6 weather app | Paid once | Money, and nothing else. It has no reason to want more |
| d | Free social network, no ads yet | Data, ads later | Users now. The advertising arrives once the audience is large enough |
| e | Department-funded homework app | Publicly funded | Nothing from the student. The department is the customer |
| f | Free flashlight, wants contacts | Data | A contact list worth more than any flashlight |

**e is not a business.** Somebody still paid — taxpayers — and the customer is the school system rather than the student. That is a genuinely different arrangement and worth naming: the app has no commercial reason to collect anything, but it does answer to whoever funded it.

## Task 10.2 · the flashlight audit

**Camera — needed.** The flash is part of the camera hardware. This is the one legitimate request.

**Contacts — unjustifiable.** A flashlight has no use for your friends' names and numbers. It is the payload.

**Location — unjustifiable** for the flashlight; valuable to an advertiser.

**Microphone — unjustifiable.** Nothing about a light needs to hear.

**Run at startup — arguable** at best; it mostly ensures the app keeps working when you are not using it.

**Network access — arguable.** Needed to fetch adverts; not needed to make light.

**a.** Refuse contacts. A well-built app carries on without them. An app that refuses to work at all has told you what it was really for.

## Task 10.3 · the terms

**1 — expected.** Standard, and set by law rather than by the company.

**2 — unacceptable.** Your photograph could appear on a billboard, in another country, forever, with no payment and no further permission.

**3 — unusual and deliberately vague.** 'Partners' is undefined, so the clause permits almost anything. Ask the class who those partners are; the honest answer is that nobody outside the company knows.

**4 — unacceptable.** A contract that one side can rewrite silently is not really an agreement.

**5 — unusual.** There is a defensible version of this for backups and legal obligations, and an indefensible version where nothing is ever deleted. The clause as written does not say which.

## Task 10.4 · what to look for

The valuable moment is part c. A class of 25 paying $2 a month raises $600 a year, which builds nothing. Free won because a very large audience paying nothing is worth more than a small audience paying something — and the only way to monetise the large audience is advertising.

Look for a refusal in part d that has a reason attached: location history, message contents, photographs of other people, anything involving someone who never agreed to the terms.

Do not push the class to a conclusion here. Students who conclude the trade is fair have made an argument, not a mistake.

## Task 10.5 · the closing question

**For:** the advertisement is relevant, the service is free, and the alternative is the same number of advertisements for things you do not want.

**Against:** an advertisement is only that well targeted because a profile of you exists — assembled without your attention, held indefinitely, and useful to anyone who buys or steals it. Usefulness is the evidence of the surveillance, not a defense against it.

Accept either side. Award the mark for whether the student has named the profile as the thing being traded.

AFTER THE LESSON

# Handover note

LEAVE THIS ON THE DESK

90 SECONDS

Photocopy this page or fill it in directly. Ninety seconds of your time saves the regular teacher a great deal of guesswork.

| SUBSTITUTE TEACHER | CLASS | DATE | PERIOD |
|---|---|---|---|
| | | | |

## Lesson used

CHECK ONE

- ☐ 01 · Clean the Data
- ☐ 02 · Charts That Lie
- ☐ 03 · Databases by Hand
- ☐ 04 · Input, Process, Output
- ☐ 05 · Making It Smaller
- ☐ 06 · Secret Writing
- ☐ 07 · Design for a Real User
- ☐ 08 · Break It Into Blocks
- ☐ 09 · Planning a Project
- ☐ 10 · What a Free App Actually Costs

### How far did the class get?

- ☐ Everyone finished the main tasks
- ☐ Most finished; some are partway
- ☐ We got through about half
- ☐ We did not get far — see notes

### Sheets

- ☐ Collected and left on the desk
- ☐ Students kept them
- ☐ Graded against the answer key

## Anything the regular teacher should know

NOTES

## Students to mention · for any reason, good or otherwise

NAMES

### Thank you

Walking into an unfamiliar subject is not easy. Leaving a clear note behind is the part that makes the next teacher’s day work, and it is genuinely appreciated.

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## Curriculum alignment

CSTA K–12 Computer Science Standards, levels 2 and 3A. Across the ten lessons this volume touches:

- Acquiring, validating and interpreting data
- Representing data honestly in charts and visualizations
- Structuring data in related tables and querying it
- Digital systems: inputs, processes, outputs, sensors and actuators
- Compression and the representation of images
- Data security, encryption and keys
- User-centered design, criteria and evaluation
- Decomposition, modularity and reuse
- Project planning, dependencies and risk
- Privacy, consent and the economics of free software

Individual lesson teacher pages name the specific focus for that lesson.

## The rest of the series

### Volume 1 · Foundations

Algorithms, binary, trace tables, flowcharts, cyber attacks, networks, sorting, debugging, AI and digital footprints.

### Volume 3 · Machines and Meaning

Lists, Boolean logic, hardware, files, requirements, machine learning, DNS, search ranking, control systems and copyright.

### Volume 4 · Media, Evidence and Impact

Images and sound as data, stacks and queues, digital forensics, version control, network design, accessibility, algorithmic decisions, IoT security and automation.

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Physical Computing with the Raspberry Pi 5 · AI Literacy for Secondary Students · and more.

## Found an error, or want a lesson on something else?

Feedback is genuinely read. If a lesson did not land with your class, or you need a sub plan on a topic that is not here, get in touch through digitalvector.com.au and it will be considered for the next update.

EMERGENCY SUB PLANS PACK

# Digital Technologies · Grades 7–10

Volume Two · ten lessons · no preparation · no computers required